

Jie Zhang

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Skills

- C++



- C#



- JavaScript



Dynamic Software Engineering Manager with over a decade at UFT development, specializing in C++, C#, and JavaScript. Proven track record in leading R&D teams to deliver innovative solutions and enhance engineering quality. Strong problem-solver with exceptional cross-cultural communication skills, driving product success in global projects.

Work History

2014-07 -

2025-06

Software Engineering Manager

HP/HPE/MicroFocus/OpenText, Shanghai

Senior software engineer and technical team lead with over a decade of experience building large-scale enterprise platforms, including as a core developer of the flagship test automation product UFT/QTP. Skilled in C++, C#, JavaScript, Docker, and browser extension development, with proven ability to adapt quickly to new technologies and drive product innovation.

Led multiple core R&D teams, delivering complex features on time while improving engineering quality and team efficiency. Experienced in managing global projects with distributed teams across the U.S., Israel, Europe, and Asia, with strong cross-cultural communication and collaboration skills.

Proactive problem-solver with a strong product mindset, capable of identifying technical risks early and translating ideas into real-world solutions.

patents:

Automated Identification of Input Elements in a GUI -
Peng-Ji Yin, Cheng Hua, Jie Zhang

Recommending Programmatic Descriptions for Test
Objects - Peng-Ji Yin, Cheng Hua, Jie Zhang

projects: [https://github.com/sagibrant/resume
/blob/master/projects.md](https://github.com/sagibrant/resume/blob/master/projects.md)

resume: [https://github.com/sagibrant/resume
/blob/master/README.MD](https://github.com/sagibrant/resume/blob/master/README.MD)

中文简历: <https://github.com/sagibrant/resume>

Education

2014-06	Master of Science <i>East China Normal University - Shanghai</i>
2011-06	Bachelor of Science <i>East China Normal University - Shanghai</i>

Projects

UFT OCR Improvement (Oct 2024 - May 2025)

Legacy OCR engines such as ABBYY and Tesseract frequently underperformed in on-premises environments, falling short of the accuracy needed for reliable AI automation. Additionally, many customers were unable to use cloud-based OCR services due to privacy constraints or infrastructure limitations.

To address these challenges, our team evaluated five potential solutions—including EasyOCR and enhanced Tesseract variants. Among them, the initial proof of concept (PoC) for a Hybrid Text Identification approach was initially rejected due to slow performance (~20 seconds per scan) and lack of word segmentation.

After I joined the initiative, I optimized the algorithm to reduce the processing time from ~20s to ~2s, while also enhancing the extraction logic to include word boundary segmentation. As a result, my PoC was selected and successfully implemented.

The Hybrid Text Identification solution intelligently fuses OCR results with DOM-based web text extraction. Key challenges included handling complex styles, nested iframe structures, and ensuring minimal performance impact during test execution. This hybrid method significantly outperformed traditional engines,

achieving up to 100% accuracy on web-rendered text—substantially improving automation reliability in secure, offline environments.

Skills: C++ and JavaScript

UFT Cloud Browser (Jan 2023 - Dec 2023)

Following the sunset of the StormRunner Functional (SRF) platform, market demand for a cloud-based browser automation solution remained strong. However, due to tight budget and resource constraints, stakeholders sought a lightweight alternative with minimal development overhead.

As the technical lead, I architected a lean Cloud Browser solution in collaboration with the Digital Lab team, achieving rapid delivery with strategic reuse and design simplification:

1. Leveraged existing Digital Lab infrastructure and UFT browser extensions to minimize new implementation overhead.
2. Redesigned the communication layer using WebSocket-based messaging, replacing complex legacy pipelines with a streamlined, maintainable protocol.
3. Reimplemented key automation features using browser extension APIs, eliminating dependencies on Windows-native APIs for improved cross-environment support.
4. Established agile co-development practices, including weekly syncs and daily hands-on code reviews with the partner team.

This effort resulted in a fully functional UFT Cloud Browser released in just 4 months with only 4 developers, dramatically outperforming the prior SRF initiative that required 3 years and ~30 engineers—demonstrating our engineering agility, architectural foresight, and ability to deliver high-value solutions under tight constraints.

Skills: C++, C# and JavaScript

UFT Parallel Execution (Jan 2018 - Jul 2018)

Historically, UFT (Unified Functional Testing) did not support parallel test execution—a major bottleneck for customers aiming to accelerate their automation pipelines. Despite UFT's minimal memory (sub-200MB) and CPU footprint, each test instance required a separate machine due to architectural limitations.

This gap became critical when pre-sales teams reported losing deals to competitors who could run 10+ mobile tests in parallel on a single laptop.

Technical Challenge:

UFT's core consists of 4 million lines of legacy code, heavily reliant on singleton COM objects and assumptions of a single instance per machine. Many modules were not designed to support concurrent execution contexts or isolation.

My Contributions:

Leveraged prior experience from UFT Cloud Execution on SRF to develop a proof-of-concept for UFT Mobile Parallel Execution.

Refactored legacy components, removing unnecessary dependencies while isolating the mobile execution logic into a standalone runner.

Enabled 10+ mobile test instances to run concurrently in a single user session, maintaining speed, reliability, and resource efficiency.

Outcome & Impact:

The POC received the recognition from the Direct manager and shifted product strategy: PM reprioritized the roadmap to focus on parallel execution.

Led the team in launching the UFT ParallelRunner, later extended to UFT Web add-in and Java add-in.

Empowered pre-sales to successfully compete in critical deals.

QA teams experienced significant test cycle reduction,

dramatically improving CI/CD throughput.
Promoted to Manager I in recognition of this
high-impact delivery.

Skills: C++, C#

More Projects:

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/projects.md](https://github.com/sagibrant/resume/blob/master/projects.md)